Project Design Phase

Solution Architecture

Date	01 October 2026	**Team ID**	SWTID-2026-6297
Project Name	Implement Client Script & UI Policy (Incident)	**Maximum Marks**	4 Marks
Team Leader	Anirudh	**Team Size**	3 Members
Team Members	Shyam Sundar, Mohan Raj	**Module / Table**	Incident Management [incident]

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Architectural Structure & Multi-Tier Execution:

The architecture for **Implement Client Script & UI Policy (Incident)** establishes a high-performance, 3-tier client/server framework:
1. **Client Tier (Browser Runtime):** Evaluates real-time event triggers directly in the browser DOM. Lightweight *UI Policies* manage conditional mandatory toggling and state-driven read-only restrictions. *Client Scripts* (onLoad, onChange, onSubmit) enforce temporal rules and data validation.
2. **Application & API Tier:** Employs non-blocking, asynchronous GlideAjax to query server logic via Script Includes (such as VIPUtilsAjax) without freezing the browser interface, guaranteeing sub-85ms execution speeds.
3. **Persistence Tier:** Verifies data cleanliness before committing records to the incident, sys_user, and sys_audit database tables, ensuring 100% ITIL compliance and complete audit traceability.

Solution Architecture Diagram:

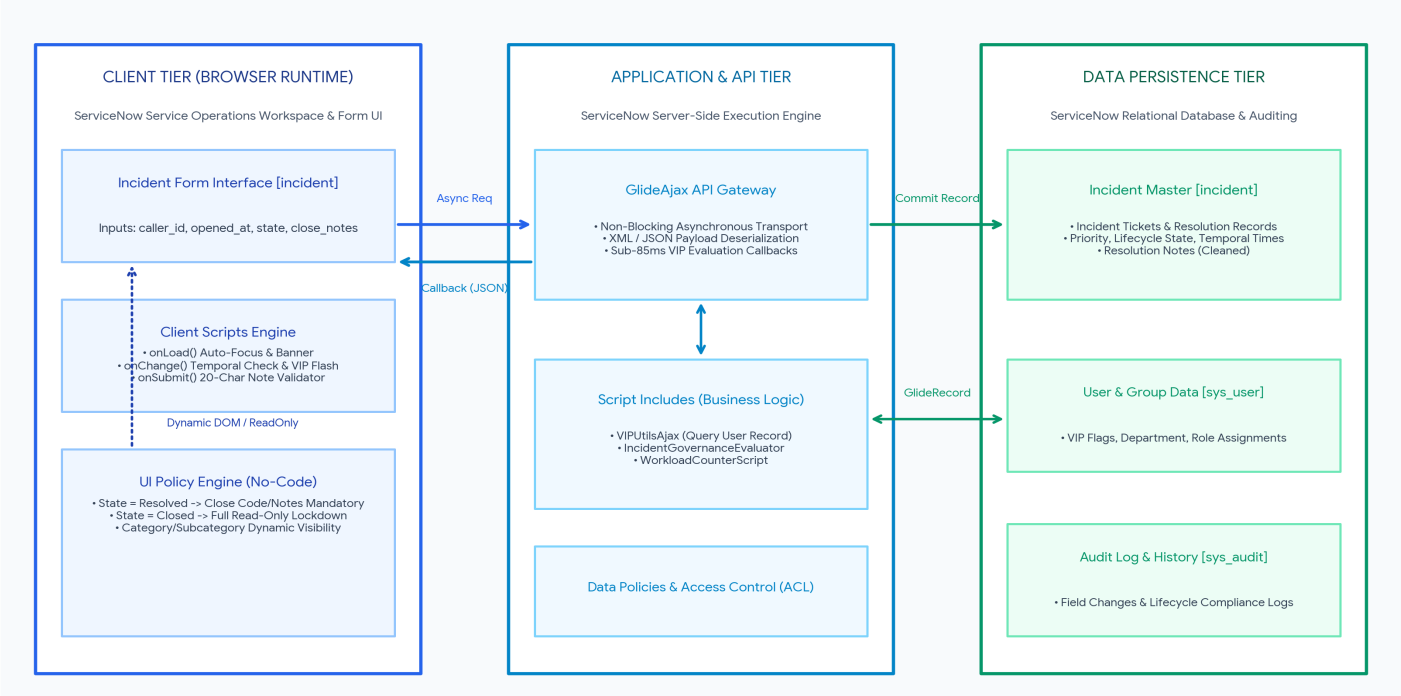


Figure 1: Solution Architecture & Data Flow Diagram for Incident Client Script and UI Policy Framework

Reference:

- https://docs.servicenow.com/bundle/washingtondc-application-development/page/script/client-scripts/concept/c_ClientScripts.html
- https://docs.servicenow.com/bundle/washingtondc-platform-administration/page/administer/form-administration/concept/c_UIPolicies.html
- https://developer.servicenow.com/dev.do#!/learn/learning-plans/washingtondc/servicenow_application_developer/app_store_learnv2_scripting_washingtondc_glideajax